

Light at the end of the tunnel...

X17 (2017) — English translation

In an isothermal atmosphere, the air is in equilibrium. Molar mass of air $\mu = 29$ g/mol, atmospheric temperature $T = 300$ K. Consider the free fall acceleration g known and constant. Also assume that the atmospheric pressure p_0 at the Earth's surface is known.

A1	Derive a formula that allows you to determine the air pressure at any height h relative to the level of the Earth's surface. Express your answer in terms of p_0 , μ , g , R , T and h .	0.50
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A2	Determine the height H at which the pressure changes e times. Express your answer using R , T , μ and g .	0.50
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The density distribution in the earth's crust is such that the acceleration due to gravity $g = 9.8$ m/s² is constant. The radius of the Earth is $r_0 = 6370$ km. The refractive index of light in air is $n = 1 + \Delta n$, where $\Delta n \ll 1$. At the surface of the Earth $n_0 = 1.000292$. Over a wide pressure range $\Delta n \sim p$.

A3	At what depth h should a tunnel be dug along the Earth's surface so that light circulates in it in a closed orbit? Consider that $h \ll r_0$. Express your answer in terms of R , T , μ , g , r_0 and n_0 . Calculate the resulting value h .	4.00
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Source: <https://pho.rs/p/308>